Quiz: Retrieval-Augmented Generation (RAG)

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Score: 20 / 20

Section A: Multiple Choice Questions (1 mark each)

1. **What is the main purpose of Retrieval-Augmented Generation (RAG)?**
   A. To compress large datasets
   B. To combine document retrieval with language generation
   C. To train a neural network from scratch
   D. To replace vector databases

   B. To combine document retrieval with language generation

2. **In a RAG system, what is the role of a vector database?**
   A. Store trained models
   B. Store embeddings for similarity search
   C. Generate answers
   D. Train transformers

   B. Store embeddings for similarity search

3. **What does “chunking” refer to in RAG pipelines?**
   A. Dividing a document into smaller sections
   B. Encrypting documents
   C. Compressing text data
   D. Converting text into images

A. Dividing a document into smaller sections

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4. **Which method is commonly used to convert text chunks into vectors?**
   A. Tokenization
   B. Embeddings
   C. Sorting
   D. Indexing

B. Embeddings

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5. **Which component generates the final response in a RAG system?**
   A. Vector database
   B. Embedding model
   C. Large Language Model
   D. Data loader

   C. Large Language Model

Section B: True or False (1 mark each)

6. RAG helps LLMs use external knowledge sources.
True

7. Chunking large documents improves retrieval accuracy.
True

8. Context injection means adding retrieved information into the prompt.
True

9. In RAG, the LLM directly searches PDFs without embeddings.
False

10. Evaluating a RAG system only requires checking model accuracy.
False

Section C: Short Answer (2 marks each)

11. Write the basic workflow of a RAG system.

Answer:

Basic workflow:

1. Load documents (PDFs, text, etc.)
2. Chunk the documents into smaller parts
3. Convert chunks into embeddings using an embedding model
4. Store embeddings in a vector database
5. User asks a query
6. Convert query into embedding
7. Retrieve similar chunks from vector DB
8. Inject retrieved context into prompt
9. LLM generates the final answer

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12. Why is chunk size important in RAG pipelines?

Answer:

Chunk size determines how much text is stored in each embedding.

- Too large: retrieval becomes less precise.
- Too small: context may be incomplete.

Proper chunk size improves retrieval accuracy and response quality.

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13. What is context injection and why is it useful?

Answer:

Context injection means adding retrieved document information into the prompt before sending it to the LLM.

It is useful because:

- It provides relevant knowledge to the model.
- Reduces hallucinations.
- Improves accuracy of answers.

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14. Name two evaluation metrics or considerations for RAG systems.

Answer:

1. Retrieval relevance (are the retrieved documents correct?)
2. Answer accuracy / faithfulness (does the answer match the retrieved context?)

Other examples: latency, hallucination rate.

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15. Give one real-world application of RAG.

Answer: Customer support chatbots that retrieve answers from company documents, FAQs, or knowledge bases to generate accurate responses.

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